

1 OBSERVATION

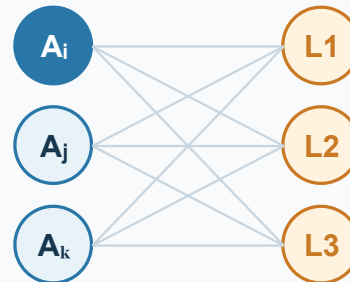
$$\mathbf{o}_i \in \mathbb{R}^{18}$$

velocity + position (4)
landmark offsets (6)
teammate offsets (4)
communication (4)

$$\mathbf{X}_i = [0, \Delta p_j, \Delta p_k, \Delta \ell_1, \Delta \ell_2, \Delta \ell_3]$$

from local relative positions

2 BIPARTITE GRAPH



$B_{uv} = 1$ for A–L edges; 0 otherwise

$$d_{uv} = \|\mathbf{X}_i[u] - \mathbf{X}_i[v]\|_2$$

$$\mathbf{W}_{uv} = \mathbf{B}_{uv} \exp(-d_{uv}^2 / 2\sigma^2)$$

$$\sigma = 0.6$$

4 POLICY INPUT

$$[\mathbf{o}_i; \mathbf{h}_i] \in \mathbb{R}^{21}$$

18 raw + 3 HKS

MADDPG actor π_i · two 64-unit ReLU layers

5 movement-action logits

3 HEAT-KERNEL FEATURES

$$\mathbf{D}_{uu} = \sum_v \mathbf{W}_{uv}$$

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$$

$$\mathbf{h}_i(\tau) = [\exp(-\tau \mathbf{L})]_{ii}$$

$\tau \in \{0.5, 1, 2\}$ · three focal returns

CTDE INTERFACE

Replay: raw observations + actions

Critic: raw joint input (69-D)

Target actor: same HKS on next observations